

Learning Patterns

Supervised Tasks and Adaptation

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Recap

Today: **keep it short!**

The Language Modeling Problem

Language Model

A Language Model (LM) estimates the probability of pieces of text. Given a sequence of text $w_1, w_2, \dots, w_S$, it answers the question:

What is $P(w_1, w_2, \dots, w_S)$?

How to compute P ?

- Count-based (count): n -gram
- Neural method (learn): Transformer
 - Masked Language Modeling (encoder-only, BERT)
 - Causal Language Modeling (decoder-only, GPT)

→ encode:

- grammaticality
- semantic plausability
- stylistic consistency
- knowledge (?)

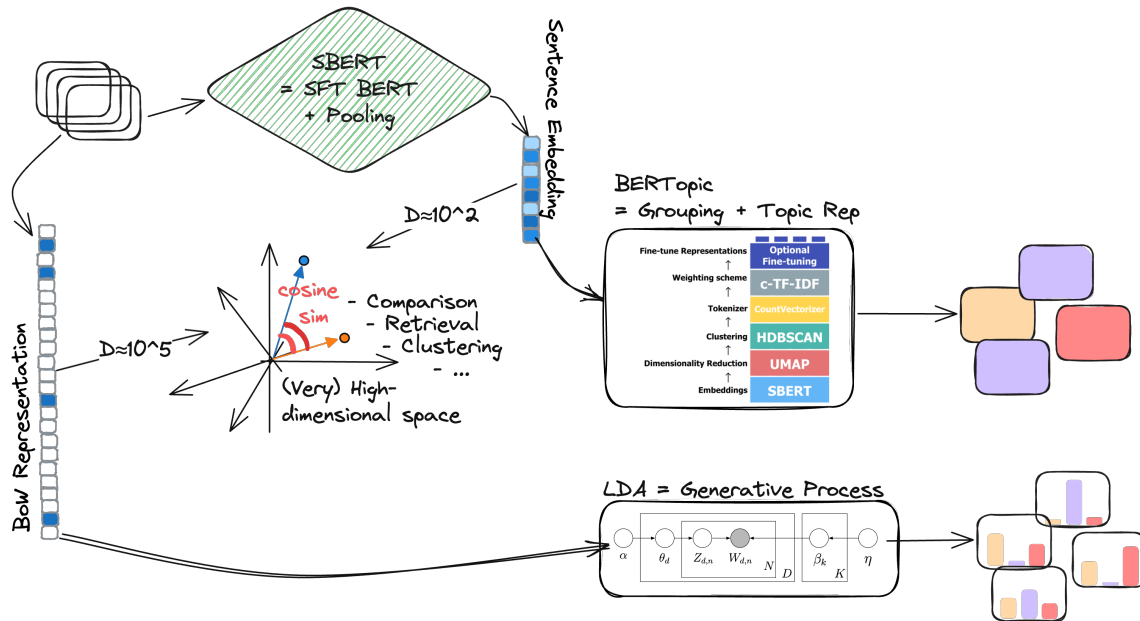
We've seen what *is* LM = probability estimation, what it can encode, and how to achieve it:

- count-based

- neural: transformers

we've started to use it to obtain words/tokens representation → whole documents!

Document Representation & Unsupervised Tasks



1. Document representation methods
2. Applied to topic modeling

but mainly: = **unsupervised** task

Supervised Learning

Today: **supervised learning** -> what's that??

Introduction

Supervised Learning

Goal

Learn patterns that map *inputs* (texts, sentences, etc.) to *outputs* (labels, categories, values) based on *annotated examples*.

What you need?

- **Training data:** documents annotated with *labels* ('ground-truth')
- A **model** that can make predictions
- A **loss function** that measures the model performance
- An **optimisation process** that nudge the model towards lower loss

→ The model will '*learn*' from examples (and corrections), improving its predictions through training.

Learn from the data w/ KNOWN LABELS

- Some data + some annotation
- A loss function
- Minimize it
- Trained model!

→ Contrast with **unsupervised learning** (seen before): no labels, the model discovers structure by itself (e.g. clustering or topic modeling).

vs: In supervised learning, we explicitly tell the model what the "right answer" is.

Supervised Learning NLP Tasks

- **Classification:** Assign a *label* to an entire text or sentence
 - *Sentiment analysis* (positive / negative reviews)
 - *Topic classification* (political / cultural / literary texts)
 - *Spam detection, author identification, genre prediction*
- **Regression:** Predict a *continuous value*
 - Predicting a *readability score* or *publication year* from a text

...

- **Sequence Labelling:** Assign a label to *tokens* in a sequence (classification)
 - *Named Entity Recognition (NER)*: tagging names of people, places, organizations
 - *Part-of-Speech tagging*
- **Question Answering (QA)**
 - The model identifies an *answer span* in a text based on a given question
- **Other tasks**
 - *Text summarization, machine translation, paraphrase detection*
(still supervised — but outputs are more complex)

Most basic supervised tasks in machine learning:

- classification: assign a categorical label
- regression: predict continuous variable

...

kinda NLP-specific variations

Idealised Supervised NLP Workflow

1. Define the task
2. Collect and annotate data
3. Prepare the data

...

4. Train the model
5. Evaluate and interpret results
6. Analyze outputs and errors
7. (Deploy?)

/!\ NEVER THAT LINEAR AND CLEAN!!!! /!\

1. **Define the task**
 - e.g., sentiment classification, genre identification, NER
2. **Collect and annotate data**
 - Texts + corresponding labels

- Ensure balance, coverage, and label quality

3. Prepare the data

- Split into *train* / *validation* / *test*
 - Tokenize or embed depending on approach
-

In the hands-on I have already more or less done these steps

4. Train the model

- Representation + classifier → fit lightweight model
- Fine-tuning → adapt pretrained model on task

5. Evaluate and interpret results

- Metrics: accuracy, F1-score, confusion matrix
- Examine misclassifications, embedding spaces, attention patterns

6. Deploy or analyze

- Use model predictions for downstream research questions
- Visualize embeddings, explore cultural/semantic trends, etc.

Main Approaches in modern NLP

Representation + standard ML

Approach

Use *precomputed text embeddings* as numerical representations of texts, then train a *standard ML model* on top.

1. Embed text

- BoW, TF-IDF, Word2Vec, Sentence-BERT, ...

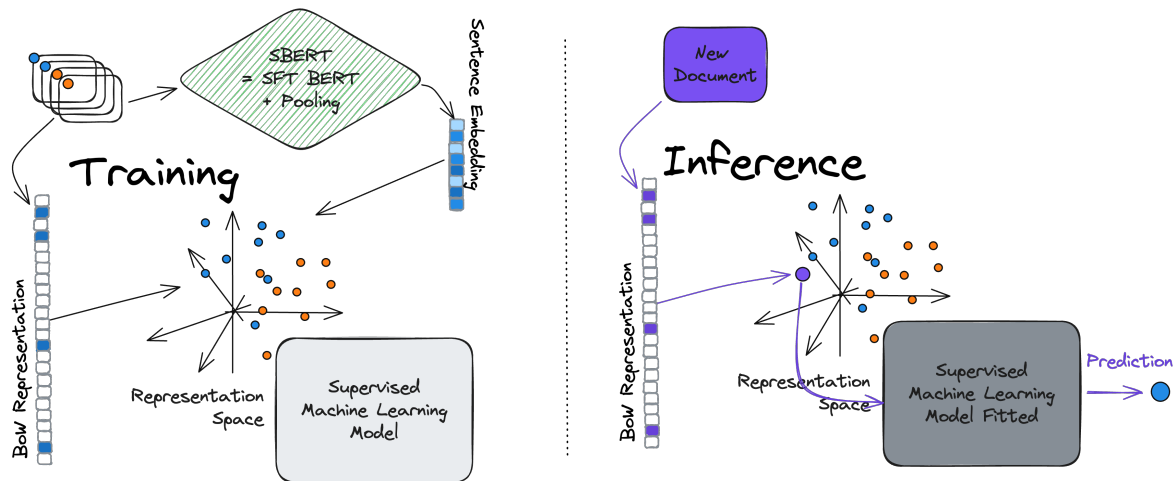
2. Train a standard ML model on the labelled embeddings

- Logistic Regression, SVM, Random Forest, MLP, ...

3. Predict in the *embedding space*

- new document → frozen embedding model → trained ML model → prediction

Rep+ML Overview



Supervised Fine-tuning

💡 Approach

Adapt a *pretrained transformer model* end-to-end to a specific downstream task by *updating its parameters* through training on labeled data.

1. Choose a **pretrained model** & add a **task-specific head**

- pre-trained LM already knows language structure (remember LM lecture)
- add a task-specific (classification, regression, ...) head on top of LM

2. **Fine-tune** the model

- update the model's weights during training thanks to the labeled data

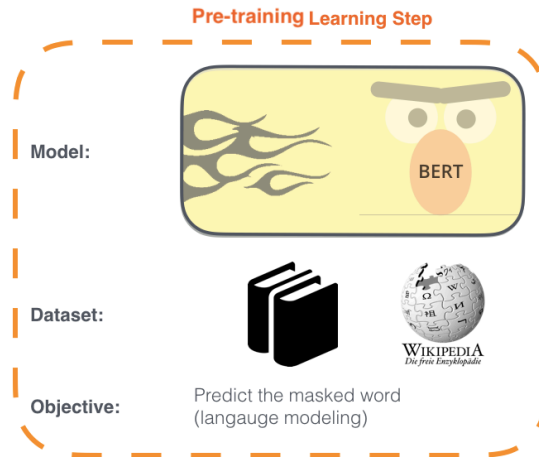
3. **Predict** directly from text

- text → fine-tuned LM → prediction

SFT Overview

1 - **Pre-training** training on large amounts of text (books, wikipedia..etc).

The model is trained on a certain task that enables it to grasp patterns in language. By the end of the training process, BERT has language-processing abilities capable of empowering many models we later need to build and train in a supervised way.



2 - **Supervised** training on a specific task with a labeled dataset.

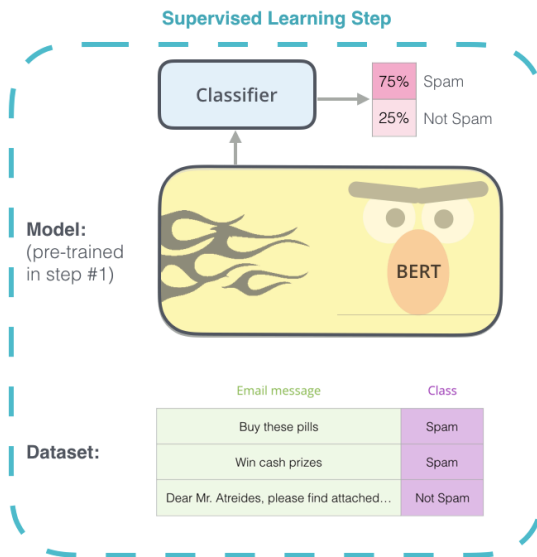


Figure 1: Two steps of BERT development (extracted and modified from (Alammar 2018)).

Summary

Takeaways

Representation + ML

Freeze embeddings, train small model
Fast, lightweight
Works with few labels
Easier to interpret & reuse
Strong for exploratory DH tasks

Fine-Tuning

Adapt all model weights
Higher compute cost
Needs more data
Task-specific, less interpretable
Best for high-performance NLP

💡 Key point

Representation + ML: modular, interpretable, resource-light

Fine-Tuning: powerful, task-adaptive, but resource-intensive

Keep in mind

- ‘*Ground-truth*’ might not be absolute
 - Model choice reflects task and data
 - Tune with intention
 - Interpret critically
-
- Both *can* rely on pretrained language models → difference is whether we **freeze or update** them.
 - Representation + Classifier → efficient, modular, transparent.
 - Fine-tuning → costly but powerful; use when accuracy matters most.
 - In Digital Humanities projects:
 - Use representation-based methods for exploratory analysis, corpus comparison, or semantic visualization.
 - Use fine-tuning when aiming for strong predictive performance on a well-defined task.

NOTES:

- labels, even more as task is subjective, OR annotators different (e.g., think toxicity detection)
- pre-trained LM can only ‘know’ about the language they were exposed to in pre-training
 - not agnostic to language
 - neither to historicity & social contexts!

Classification

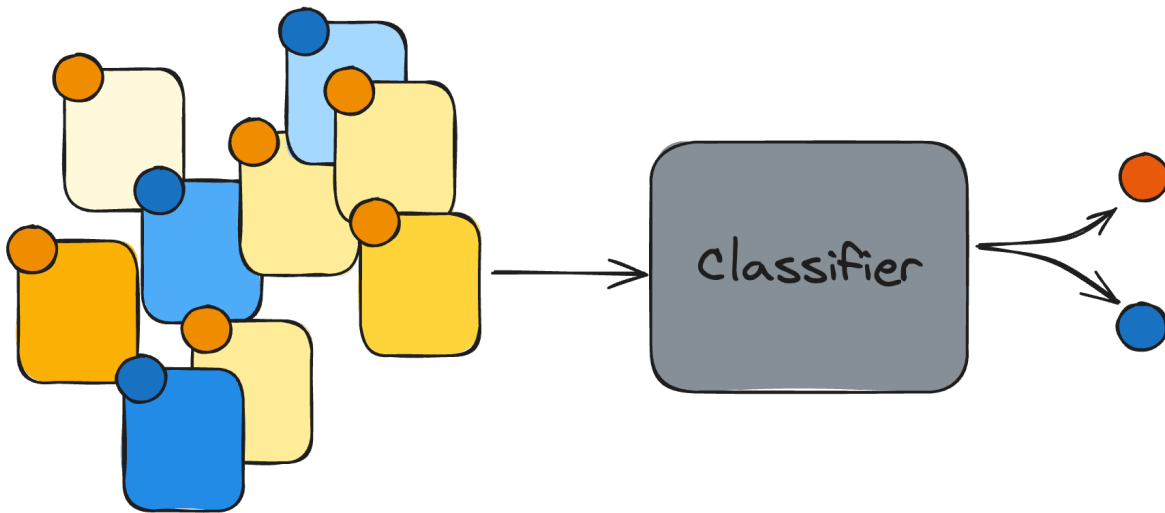
A quick *focus* (and revisions) on classification

What is a classification task?

Description

💡 Goal

Assign each input text to one of a set of predefined categories or classes.



Examples:

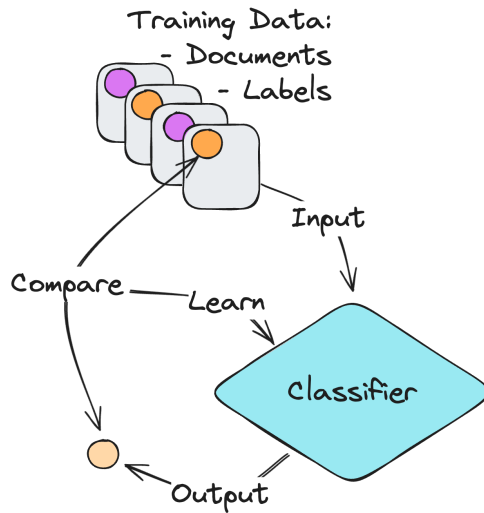
- Sentiment analysis → *positive / negative / neutral*
- Authorship attribution → *author A / author B / author C*

Key ideas

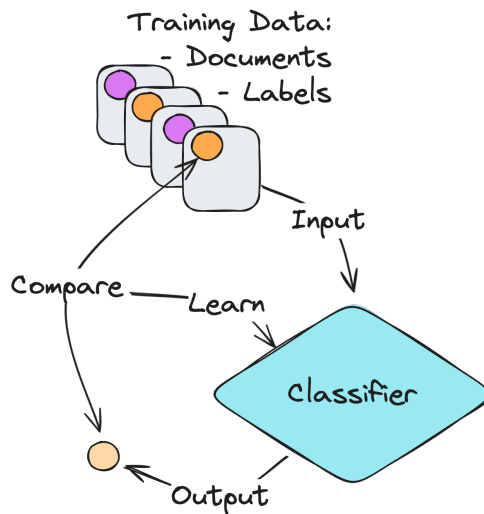
- The model outputs **discrete labels** (not continuous values)
- Often based on **probability scores** (e.g., “positive” = 0.87)
- Evaluation relies on **how well predicted labels match true labels**

Overview

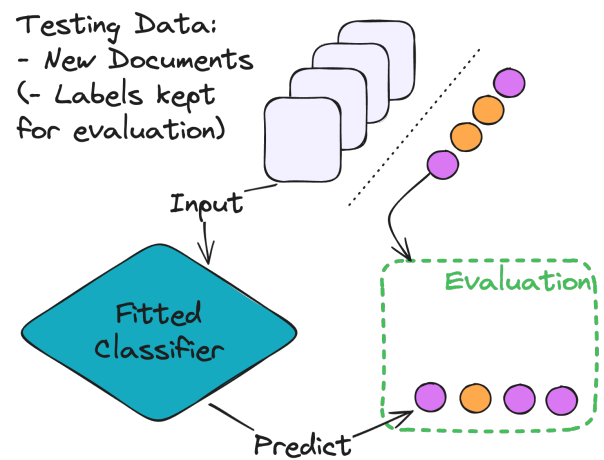
Training Stage



Training Stage



Testing Stage



In practice: that's it

...

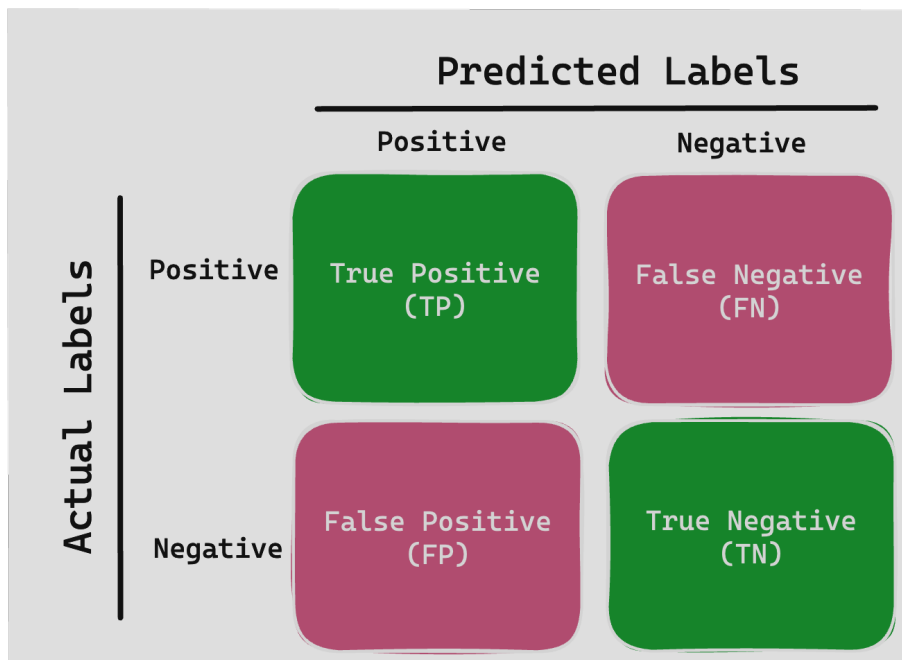
→ but how is the evaluation process???

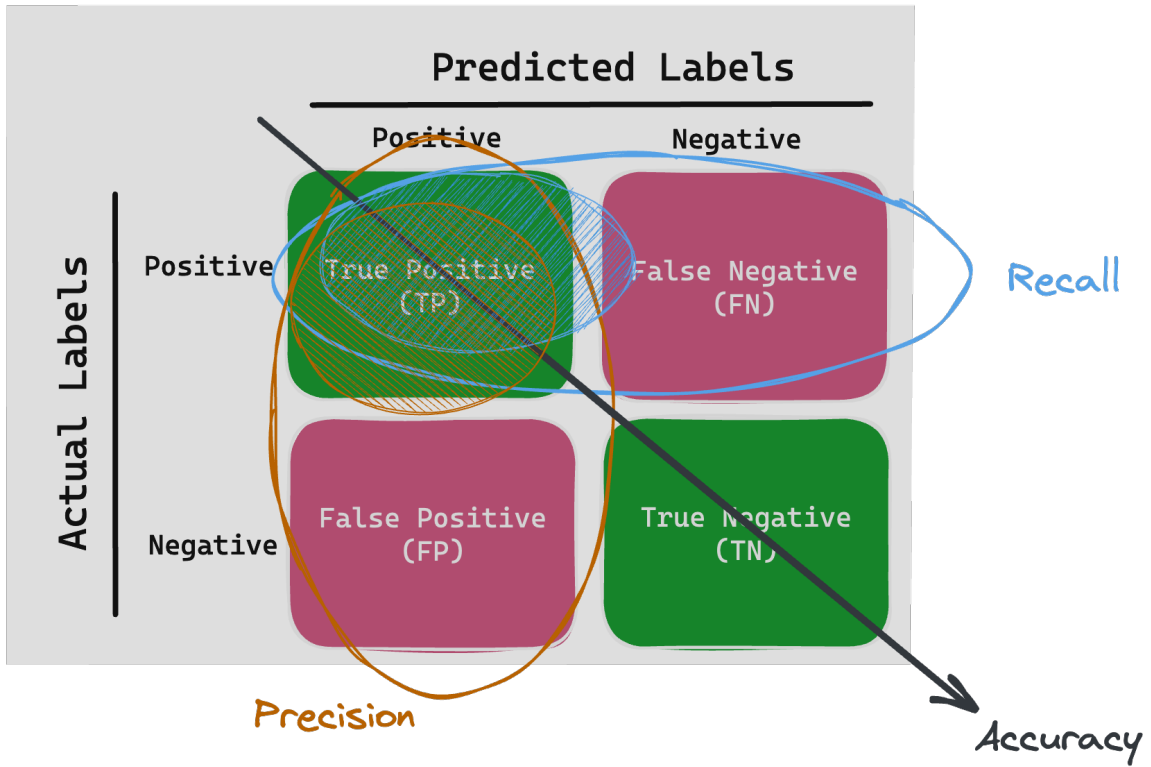
The Confusion Matrix & Performance Metrics

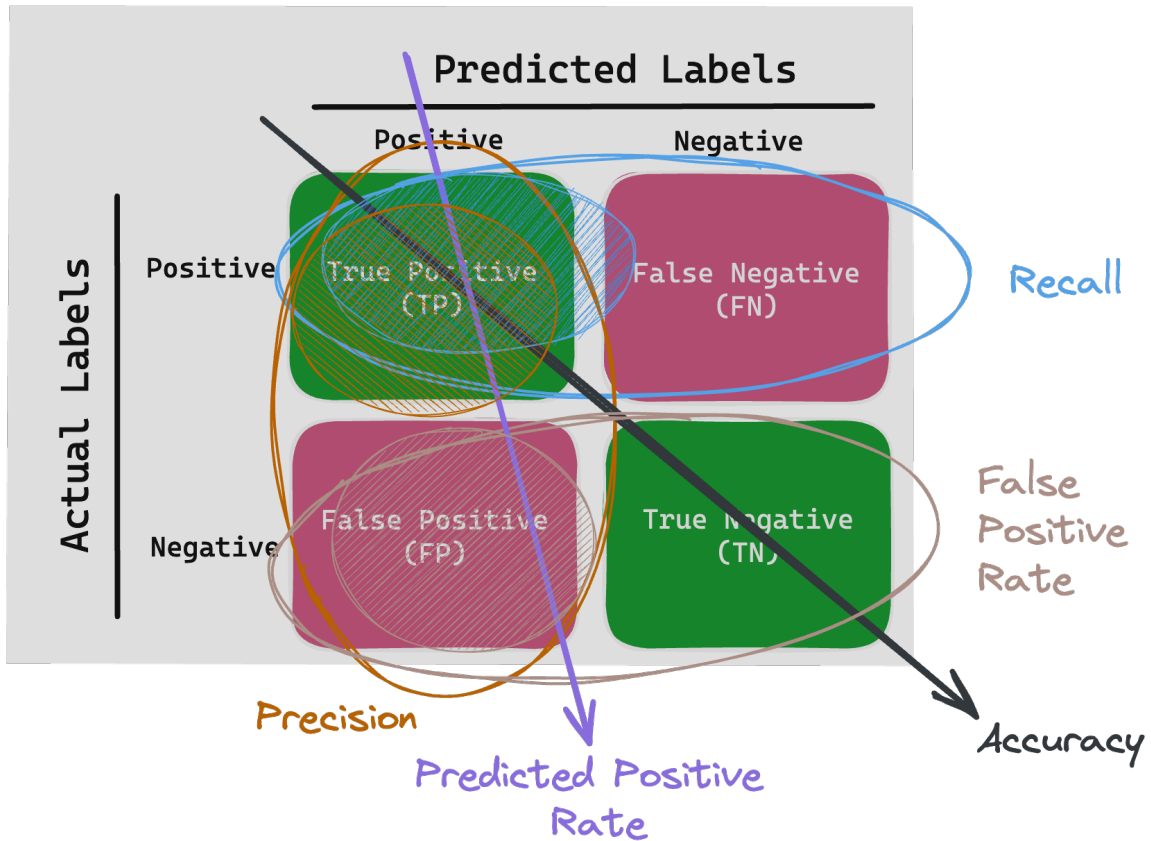
Confusion Matrix

Definition

Matrix where each cell counts examples by how the predictions compare to the ground truth.







→ Used to compute different performance metrics (see also: [wikipedia](https://en.wikipedia.org/wiki/Confusion_matrix))

Spend a little bit of time defining **each quadrants of CM**: + DRAW!

- 1st letter: how the model did — if it was succesful (T), or not (F)
- 2nd letter: actual label — positive class (P), negative class (N)

...

Used to compute many different performance indicators, that measure different *aspect of performance*.

→ Sidenote: *positive* class can sometimes be quite arbitrary (and has an importance on the scores)

Performance Metrics

Metric	Formula	Interpretation
Accuracy	$\frac{TP+TN}{TP+TN+FP+FN}$	› Overall proportion of correct predictions
Positive Predicted Value	$\frac{TP}{TP+FP}$	› How accurate the positive predictions are (=precision)
True Positive Rate	$\frac{TP}{TP+FN}$	› Coverage of actual positive sample (=recall)
F1-score	$\frac{2 \times PPV \times TPR}{PPV + TPR}$	› Harmonic mean of precision and recall
False Positive Rate	$\frac{FP}{FP+TN}$	› Proportion of incorrect positive predictions
Predicted Positive Rate	$\frac{TP+FP}{TP+TN+FP+FN}$	› Proportion of samples predicted as positive

TODO: + multi-class??

Derived from the confusion matrix — each metric highlights a different *aspect of performance*.

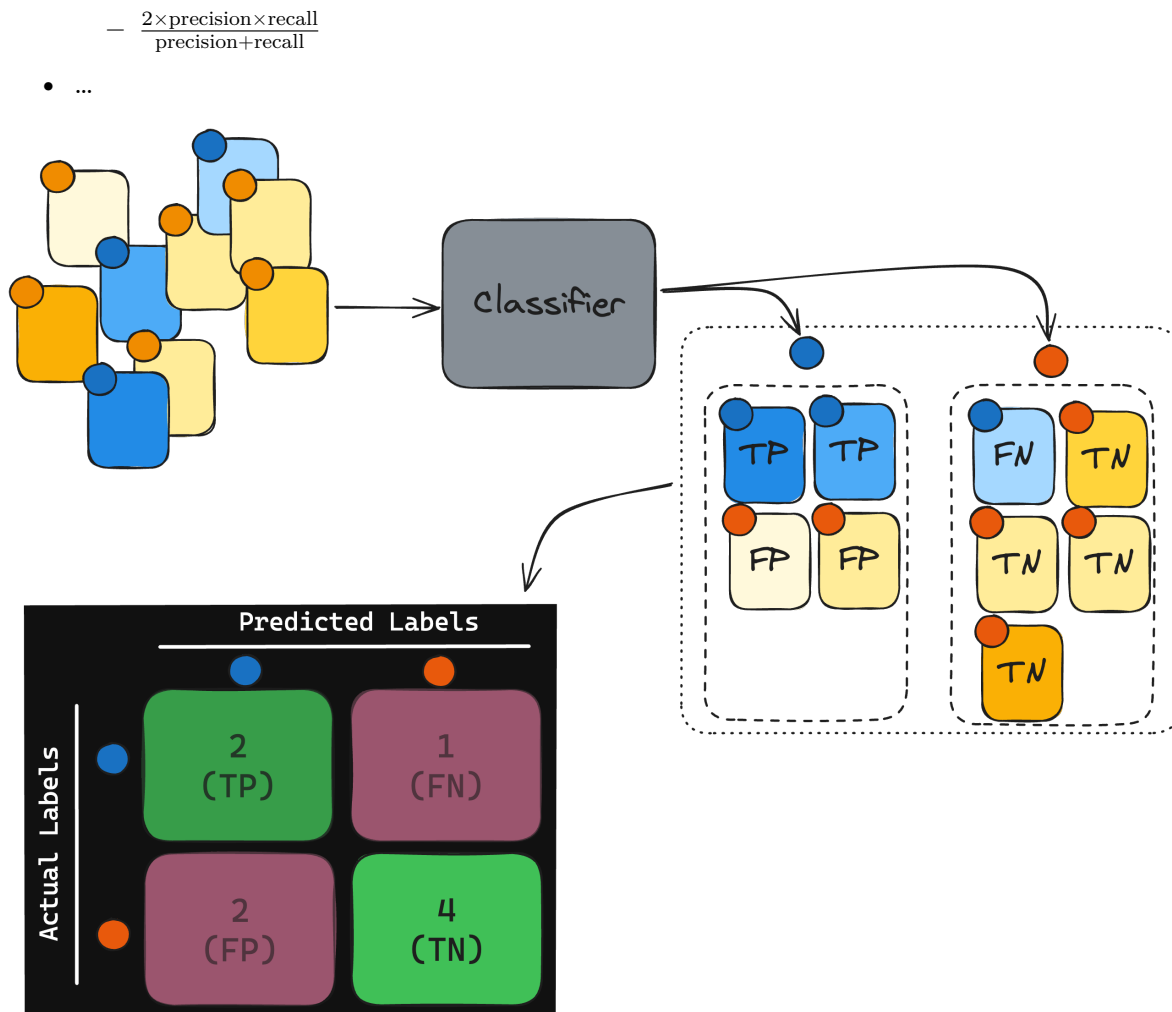
When to use what:

- Accuracy is intuitive but misleading for imbalanced datasets
- Precision/Recall highlight *different trade-offs* (e.g., minimizing false alarms vs. missing true cases)
- F1 is a balanced summary for uneven class distributions

Discuss why accuracy might look high even if the model rarely detects the minority class.

Quick Example

- Accuracy
 - $(2+4)/(2+1+2+4) = 6/9 \approx 0.66$
- Precision
 - $2/(2+2) = 2/4 = 0.5$
- Recall
 - $2/(2+1) = 2/3 \approx 0.66$
- F1-score



- $\text{precision} = \text{TP} / (\text{TP} + \text{FP})$
- $\text{recall} = \text{TP} / (\text{TP} + \text{FN})$
- $F1 \sim 0.5714$

Is that enough?

Say I have a model that have high performance scores. Can I rest now?

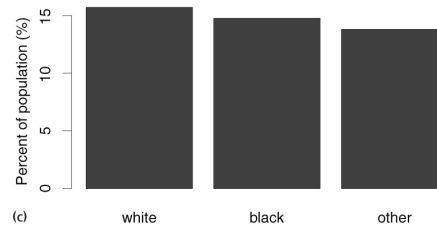
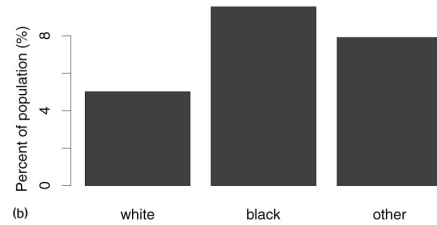
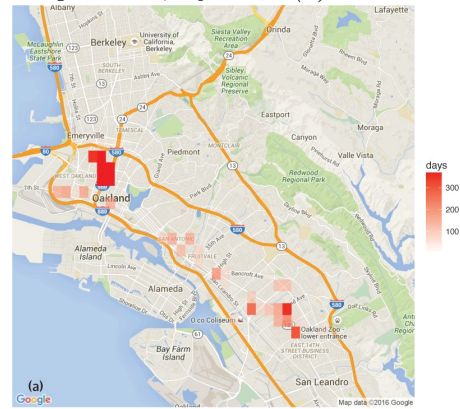
Some examples...

... reinforcing structural discriminations, through:

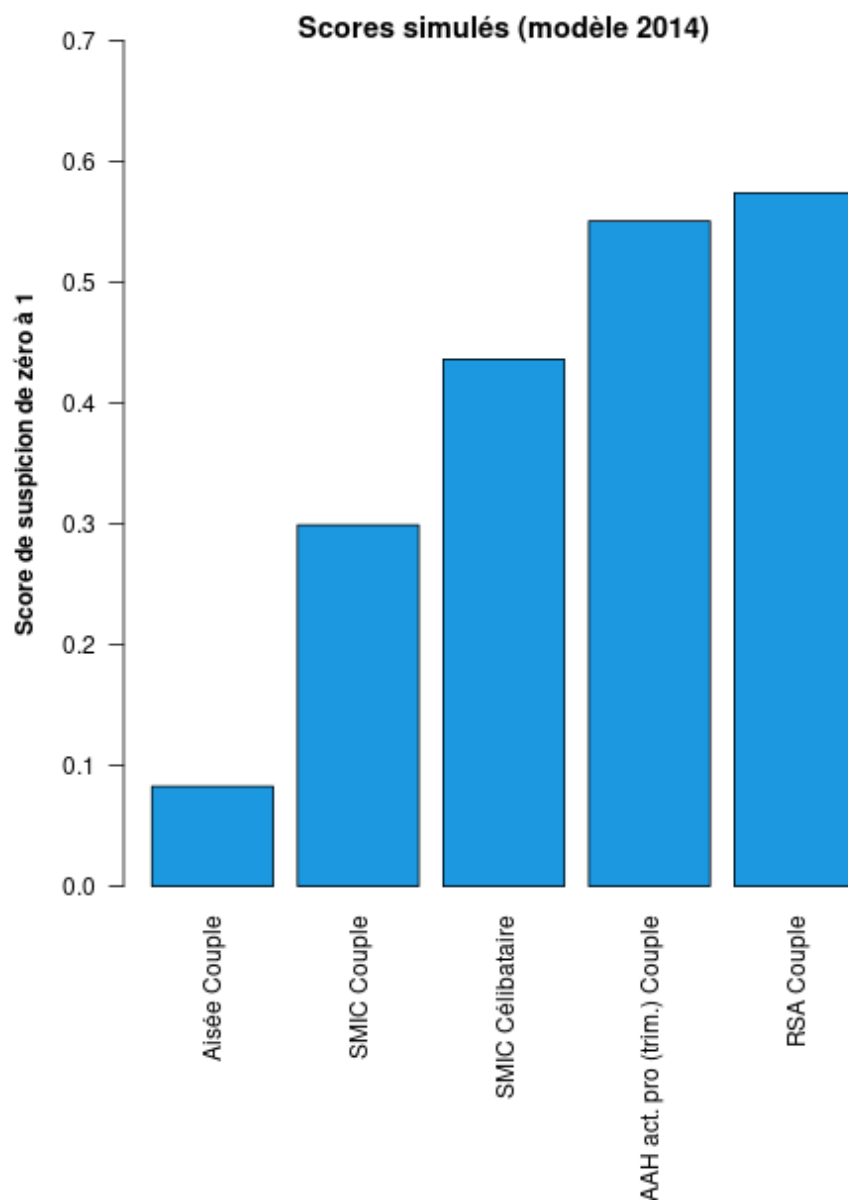
- Reported Bias & Runaway Feedback Loops

- Minorities targeting in predictive justice (Angwin et al. 2016); predictive policing (Lum and Isaac 2016; Rotaru et al. 2022)
- Latent Bias & Sampling Bias
 - Racial healthcare disparities (Obermeyer et al. 2019)
 - Gender bias in credit scoring (Telford 2019) and recruitment practices (Dastin 2022)
 - Classist biases in welfare systems (QdN 2023)
- ...

(a) Number of days with targeted policing for drug crimes in areas flagged by PredPol analysis of Oakland police data. (b) Targeted policing for drug crimes, by race. (c) Estimated drug use



by race (extracted from (Lum and Isaac 2016)).



CAF's suspicion score (extracted from (QdN 2023)).

Not so sure... even *strong* classifier, can be very biased and have **highly harmful** repercussions.

Here are some real-world examples of biased algorithms...

Of course everything is a bit intertwined... (and not limited to gender and racial biases!)

- reported bias: data is biased by real-world practices and/or historical biases (e.g. police over-patrolling minority neighborhoods)

- runaway feedback loops: reinforcement (in wrong direction = vicious circle)
- latent: Proxy variables (like cost or hospital visits) can encode structural inequalities, leading to skewed predictions.

Not sci-fi, not US only...

Fairness Metrics

so (at least in some cases) it might be good to think about fairness
and there exist metrics developed for that

Conceptual Overview

Three core concepts (Barocas, Hardt, and Narayanan 2023):

1. **Independence** — predictions are statistically independent of sensitive attributes
 - e.g., same positive rate for all groups
2. **Separation** — error rates are similar across groups, conditional on the true label
 - e.g., same false-positive / false-negative rates for each group
3. **Sufficiency** — predictions are equally reliable across groups
 - e.g., for a given predicted probability, the actual correctness is the same

Considering a minority group or non-privileged or ‘endangered’ or ...

3 CORE CONCEPTS: Independence, Separation, Sufficiency

Underlying idea

Fairness across groups

The model should perform similarly across groups.

→ compute performance metrics for different groups: let p refer to privileged group and n to non-privileged group:

$$m_p = \text{score}(\text{TP}_p, \text{FP}_p, \text{TN}_p, \text{FN}_p), \quad m_n = \text{score}(\text{TP}_n, \text{FP}_n, \text{TN}_n, \text{FN}_n)$$

→ compare them with a ratio. A “fair” model should ensure (RoT: $\epsilon \approx 0.8$):

$$\epsilon \leq \frac{m_n}{m_p} \leq 1/\epsilon$$

Fairness between groups

HERE: bias/fairness is not about low accuracy, but *really* a **systematic difference in performance across groups**

In practice: operationalize these notions as *ratios* for different performance metrics across groups.

Then, we can see if this ratio lies within a ‘fairness’ confidence interval ($\rightarrow \epsilon$)

-> We’ll see examples in the hands-on!

Describe the 5 metrics used in hands-on session (the ones that are in the `dalex` Python Package).

Practical Measures of Fairness

Name	Metric	Concept	Interpretation (Wiśniewski and Biecek 2022)
Statistical Parity	PPR	Independence (equivalent)	whether both groups have equal likelihood of being <i>predicted as positive</i>
Equal Opportunity	TPR	Separation (relaxation)	likelihood of <i>correctly</i> recognising a <i>positive</i> is equal regardless of group
Predictive Equality	FPR	Separation (relaxation)	likelihood of <i>incorrectly</i> classifying a <i>negative</i> as <i>positive</i> is equal regardless of group
Predictive Parity	PPV	Sufficiency (relaxation)	whether <i>positive</i> are equally reliable across groups
Accuracy Difference	ACC	—	whether the model <i>performance</i> is <i>consistent</i> across groups

These quantify *how performance varies across subgroups* and help identify potential bias sources.

Frame fairness as *a diagnostic, not an accusation*:

we check whether models behave differently across groups, which may reflect data imbalance or annotation bias.

Remember to interpret fairness metrics critically and contextually.

[Reminder: *based on per-group performance*]

Metric	Formula	Interpretation
Predicted Positive Rate (PPR)	$\frac{TP+FP}{TP+TN+FP+FN}$	› Proportion of samples predicted as positive
True Positive Rate (TPR)	$\frac{TP}{TP+FN}$	› Coverage of actual positive sample (=recall)
False Positive Rate (FPR)	$\frac{FP}{FP+TN}$	› Proportion of incorrect positive predictions
Positive Predicted Value	$\frac{TP}{TP+FP}$	› How accurate the positive predictions are (=precision)
Accuracy	$\frac{TP+TN}{TP+TN+FP+FN}$	› Overall proportion of correct predictions

Key Message

- Strong Model $\nRightarrow$ Fair Model
- (Fair Model $\nRightarrow$ Strong Model)

TODO!

Hands-on

Hands-On Challenge

References

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